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GeeksQuiz

- [Home](#)
- [Algorithms](#)
- [DS](#)
- [GATE](#)
- [Interview Corner](#)
- [Q&A](#)
- [C](#)
- [C++](#)
- [Java](#)
- [Books](#)
- [Contribute](#)
- [Contests](#)
- [Jobs](#)

[Array](#)

[Bit Magic](#)

[C/C++](#)

[Articles](#)

[GFacts](#)

[Linked List](#)

[MCQ](#)

[Misc](#)

[Output](#)

[String](#)

[Tree](#)

[Graph](#)

Count Inversions in an array

Inversion Count for an array indicates – how far (or close) the array is from being sorted. If array is already sorted then inversion count is 0. If array is sorted in reverse order that inversion count is the maximum.

Formally speaking, two elements $a[i]$ and $a[j]$ form an inversion if $a[i] > a[j]$ and $i < j$

Example:

The sequence 2, 4, 1, 3, 5 has three inversions (2, 1), (4, 1), (4, 3).

METHOD 1 (Simple)

For each element, count number of elements which are on right side of it and are smaller than it.

```
int getInvCount(int arr[], int n)
{
    int inv_count = 0;
    int i, j;
```

```

for(i = 0; i < n - 1; i++)
    for(j = i+1; j < n; j++)
        if(arr[i] > arr[j])
            inv_count++;

return inv_count;
}

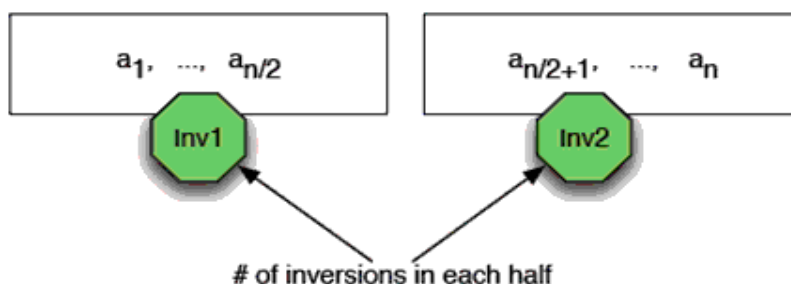
/* Driver program to test above functions */
int main(int argc, char** args)
{
    int arr[] = {1, 20, 6, 4, 5};
    printf(" Number of inversions are %d \n", getInvCount(arr, 5));
    getchar();
    return 0;
}

```

Time Complexity: $O(n^2)$

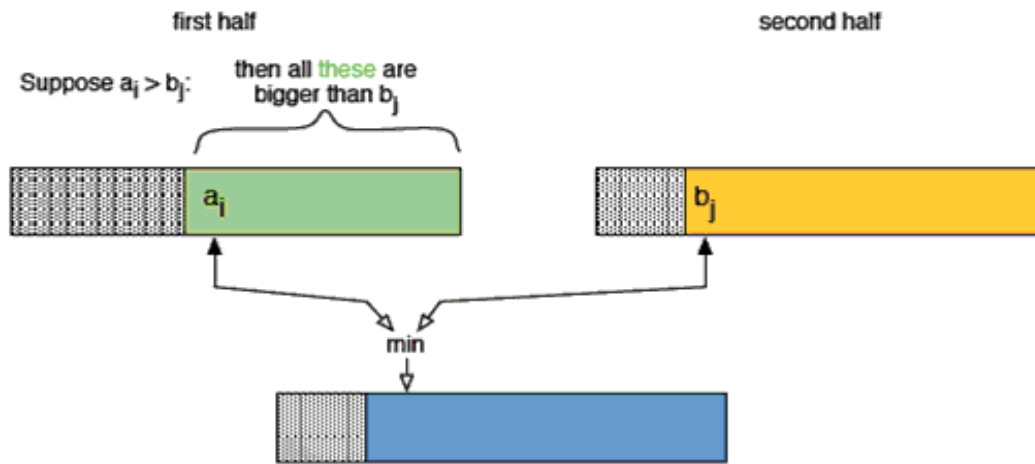
METHOD 2(Enhance Merge Sort)

Suppose we know the number of inversions in the left half and right half of the array (let be $inv1$ and $inv2$), what kinds of inversions are not accounted for in $Inv1 + Inv2$? The answer is – the inversions we have to count during the merge step. Therefore, to get number of inversions, we need to add number of inversions in left subarray, right subarray and merge().

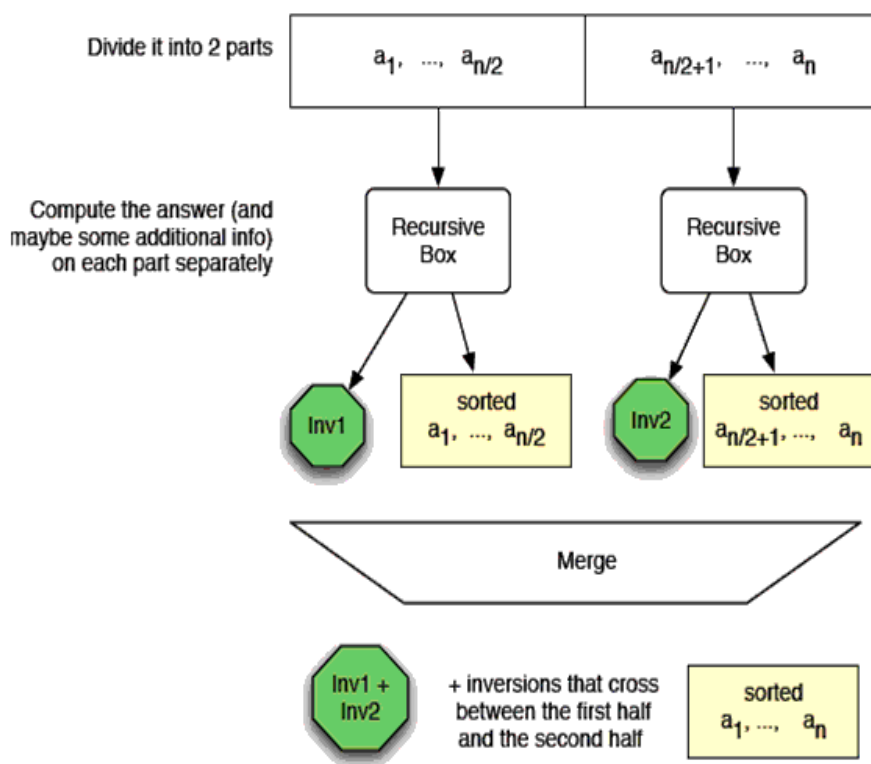


How to get number of inversions in merge()?

In merge process, let i is used for indexing left sub-array and j for right sub-array. At any step in merge(), if $a[i]$ is greater than $a[j]$, then there are $(mid - i)$ inversions. because left and right subarrays are sorted, so all the remaining elements in left-subarray ($a[i+1]$, $a[i+2]$... $a[mid]$) will be greater than $a[j]$



The complete picture:



Implementation:

```
#include <stdio.h>
#include <stdlib.h>

int _mergeSort(int arr[], int temp[], int left, int right);
int merge(int arr[], int temp[], int left, int mid, int right);

/* This function sorts the input array and returns the
   number of inversions in the array */
int mergeSort(int arr[], int array_size)
{
    int *temp = (int *)malloc(sizeof(int)*array_size);
    return _mergeSort(arr, temp, 0, array_size - 1);
}
```

```

/* An auxiliary recursive function that sorts the input array and
returns the number of inversions in the array. */
int _mergeSort(int arr[], int temp[], int left, int right)
{
    int mid, inv_count = 0;
    if (right > left)
    {
        /* Divide the array into two parts and call _mergeSortAndCountInv()
        for each of the parts */
        mid = (right + left)/2;

        /* Inversion count will be sum of inversions in left-part, right-part
        and number of inversions in merging */
        inv_count = _mergeSort(arr, temp, left, mid);
        inv_count += _mergeSort(arr, temp, mid+1, right);

        /*Merge the two parts*/
        inv_count += merge(arr, temp, left, mid+1, right);
    }
    return inv_count;
}

/* This funt merges two sorted arrays and returns inversion count in
the arrays.*/
int merge(int arr[], int temp[], int left, int mid, int right)
{
    int i, j, k;
    int inv_count = 0;

    i = left; /* i is index for left subarray*/
    j = mid; /* i is index for right subarray*/
    k = left; /* i is index for resultant merged subarray*/
    while ((i <= mid - 1) && (j <= right))
    {
        if (arr[i] <= arr[j])
        {
            temp[k++] = arr[i++];
        }
        else
        {
            temp[k++] = arr[j++];

            /*this is tricky -- see above explanation/diagram for merge()*/
            inv_count = inv_count + (mid - i);
        }
    }

    /* Copy the remaining elements of left subarray
    (if there are any) to temp*/
    while (i <= mid - 1)
        temp[k++] = arr[i++];

    /* Copy the remaining elements of right subarray
    (if there are any) to temp*/
    while (j <= right)
        temp[k++] = arr[j++];
}

```

```

/*Copy back the merged elements to original array*/
for (i=left; i <= right; i++)
    arr[i] = temp[i];

return inv_count;
}

/* Driver progra to test above functions */
int main(int argv, char** args)
{
    int arr[] = {1, 20, 6, 4, 5};
    printf(" Number of inversions are %d \n", mergeSort(arr, 5));
    getchar();
    return 0;
}

```

Note that above code modifies (or sorts) the input array. If we want to count only inversions then we need to create a copy of original array and call mergeSort() on copy.

Time Complexity: $O(n \log n)$

Algorithmic Paradigm: Divide and Conquer

References:

<http://www.cs.umd.edu/class/fall2009/cmssc451/lectures/Lec08-inversions.pdf>

<http://www.cp.eng.chula.ac.th/~piak/teaching/algo/algo2008/count-inv.htm>

Please write comments if you find any bug in the above program/algorithm or other ways to solve the same problem.



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Sachin Shenoy • 7 days ago

Assuming integers are represented using fixed length bits, here is an $O(N)$ solution to this problem

<http://ideone.com/g57O87>

We are able to achieve less than $O(N\log N)$ time complexity, as we are using idea behind a non-comparative sorting algorithm to get the count.

^ | ▾ • Reply • Share ▸



Krunal Panchal • 9 days ago

A simple code than above, using inversions global variable

code is below

```
#include <cstdlib>
#include <iostream>
```

```
using namespace std;
```

```
int b[5];
int inversions=0;
void merge(int *a,int left,int mid,int right)
{
    int i,j,k;
    i=left;
    k=left;
    j=mid+1;
    while((i<=mid)&&(j<=right))
    {
```

[see more](#)

^ | ▾ • Reply • Share ▸

**pauline** • a month ago

how can i Write a function which shifts the elements of an array (of integers) by k positions. For example, if k=5 and the array is [1 2 3 4 5 6 7 8 9] then after we callt the function on the array and k, the array will be [6 7 8 9 1 2 3 4 5].

^ | v • Reply • Share ›

**Sumeet Varma** → pauline • a month ago

Naively shifting the array always would cause $O(n)$ always.

Here is an $O(1)$ solution per each shift with just addition and modulus.

u don't actually need to shift the array.. just keep a head pointer initially at 0 position. Now say u want to shift everything by 5 position by left, then head = (head + 5) % n because our head would move 5 position by right then. Also think of array as a circular array so that whenever head exceeds n (right end of array) u wrap it around from 0 (left end of array)

also in general if after shifting u want to access say i'th element of new array. then `int element_at_position_i = a[(i+ head) % n]`

if u want to loop across the array. just start looping from `i = head` to `i = head + n`. and access element as `a[i%n]`

1 ^ | v • Reply • Share ›

**Ashish Maheshwari** • 2 months ago

In the paragraph..

How to get number of inversions in merge()?

In merge process, let i is used for indexing left sub-array and j for right sub-array. At any step in merge(), if "`a[i]` is greater than `a[j]`", then there are (mid – i) inversions. because left and right subarrays are sorted, so all the remaining elements in left-subarray (`a[i+1]`, `a[i+2]` ... `a[mid]`) will be greater than `a[j]`

There is a typo. it should be "`a[i]` is greater than `b[j]`". Please correct it :)

^ | v • Reply • Share ›

**Mahesh Kasana** • 2 months ago

Inversion finding is very easy by tree implementation.....

first of all make "invers" as a global or class member

and after complete tree creation print "inver".

```
void array :: inserttreedata(int data)
```

```
{
```

```
tree *p,*q;
```

```
p=new tree;
```

```
p->data=data;

p->count=1;

p->left=p->right=NULL;

if(root==NULL)
```

[see more](#)

^ | v • Reply • Share ›



Ryan Sun • 3 months ago

I think here should be $(mid - i + 1)$, `if(arr[j] < arr[i])`

```
else
{
temp[k++] = arr[j++];
```

```
/*this is tricky -- see above explanation/diagram for merge()*/
inv_count = inv_count + (mid - i);
```

^ | v • Reply • Share ›



mudit → Ryan Sun • a month ago

That's perfect. Here mid is actually mid + 1. See following line of code:

```
/*Merge the two parts*/
inv_count += merge(arr, temp, left, mid+1, right);
```

^ | v • Reply • Share ›



Aadil • 4 months ago

What if I want to count inversions for triplets?

^ | v • Reply • Share ›



Mr.Wonderful • 4 months ago

comments in the code need to be updated.

^ | v • Reply • Share ›



Shiva Attry • 4 months ago

Here's a c++ code to calculate number of inversion, working for several cases, not much sure though. Any suggestion?

<http://ideone.com/MWeQpx>

^ | v • Reply • Share ›



Fernando Ferreira • 4 months ago

This is my implementation in C. It's a very clean code and easier to understand.

<http://ideone.com/Z471KB>

1 ^ | v • Reply • Share ›

**Aditya Goel** • 4 months ago

without using temp (done change in G2G Standard Merge Sort code)

<http://ideone.com/XM7xFc>

^ | v • Reply • Share ›

**user** • 4 months ago

A python Implementation:

<http://ideone.com/XbQmEL>

1 ^ | v • Reply • Share ›

**Guest** • 4 months ago

whats wrong with input {6, 3, 8, 2, 5, 1, 9}.

Program's output for it is 6, it should be 11.

Please point out the bug

2 ^ | v • Reply • Share ›

**Aditya Goel** → **Guest** • 4 months ago

Yes, u r right, something is breaking

It is giving correct o/p in my code-

<http://ideone.com/XM7xFc>

^ | v • Reply • Share ›

**TJ** • 4 months ago

what should be inversion count for {5,4,2,6,7}, ideally only swapping 5 and 2 will do..so it should be 1 but above methods will return more than 1.

^ | v • Reply • Share ›

**user** → **TJ** • 4 months ago

Thats fine, you can sort this array in one swap, but not using Merge Sort. By doing some paper work you will get the clear picture. For inversions go again with problem definition. In this array, 3 inversions are there: (5,4), (5,2) and (4,2)

^ | v • Reply • Share ›

**sid** • 4 months ago

We can also count no of inversions by using the no of steps which are being executed in the insertion sort like this :-

```
int numInversions(int nArray[], int nLength)
//
int nInversions <- 0
for int i <- 0 to nLength - 1
do
j <- i - 1
while j is greater than or equal to 0 and nArray[j] > nArray[j + 1]
do
swap nArray[j] and nArray[j + 1]
```

```

swap(arr[i], arr[j]);
nInversions++;

```

```

return nInversions

```

^ | v • Reply • Share ›



Jasdeep Singh → sid • 4 months ago

your soln has $O(n^2)$ complexity for $n > 10^6$ it would definitely give tle

^ | v • Reply • Share ›



neer1304 • 4 months ago

<http://ideone.com/ds5zlu>

^ | v • Reply • Share ›



shiva krishna • 4 months ago

One question, isn't the time complexity of method one, $O(n!)$?

^ | v • Reply • Share ›



Ekta Goel → shiva krishna • 4 months ago

NO, count the number of comparisons, for the first iteration, $(n-1)$ for second $(n-2)$ for third, $(n-3)$ and so on till $i=n-1$. This is basically the summation of $(n-i-1)$ for all i from 0 to $n-1$. This comes out to be $O(n^2)$.

^ | v • Reply • Share ›



Bill Gates • 5 months ago

Nice post.

^ | v • Reply • Share ›



Guest • 5 months ago

I understand that the most inversions happen when the array is reversed, but is it possible to determine this number (using some sort of formula)?

^ | v • Reply • Share ›



Boga → Guest • 3 months ago

if the array is reversed, then every number is larger than all the numbers that are to the right side of the number. So basically it becomes

$$(n-1) + (n-2) + (n-3) + \dots + 1 + 0 = n(n+1)/2 - n = n(n-1)/2$$

^ | v • Reply • Share ›



Peter Fenwick • 6 months ago

BIT

2 ^ | v • Reply • Share ›



Deepak Rishi • 7 months ago

One can also solve this question by tweaking binary search trees. In each node keep a tab on the number of nodes that are to the right of the node. when you are inserting a node keep in mind the following : if you go left add 1 and the number of nodes to the right of the node

in mind the following . If you go left add 1 and the number of nodes to the right of the node from which you went left. If you go right , simply increment the number of nodes to the right of that node by one.

However, if you have a large dataset(especially if your tree is not complete) , you might run into the problem of memory becoming full.

2 ^ | v • Reply • Share ›



Amit Handa • 7 months ago

Anything wrong with using insertion sort.

```
int findInversions( int[] arr, int n ) {
```

```
    int invCount = 0;
```

```
    for( int i = 1; i < n; i++ ) {
```

```
        int ele = arr[i];
```

```
        int j;
```

```
        for( j = i-1; j > -1; j++ ) {
```

```
            if( arr[j] > ele ) {
```

```
                arr[i] = arr[j];
```

```
                invCount++;
```

```
            } else break;
```

```
        }
```

```
        arr[j] = ele;
```

```
    }
```

```
    return invCount;
```

```
}
```

edit : its n2.

1 ^ | v • Reply • Share ›



Guest • 8 months ago

@GeeksforGeeks the 2nd method is having some problem , check for input {8,7,6,5,4,3,2,1} giving output 10, while it should be 28

3 ^ | v • Reply • Share ›



Anurag Singh ➔ Guest • 7 months ago

You would have done following:

```
int arr[] = {8,7,6,5,4,3,2,1};
```

```
printf(" Number of inversions are %d \n", mergeSort(arr, 5));
```

This gives answer 10

In mergeSort call, change 5 to 8, as your array size is 8 (Not 5)

Correct line is:

```
printf(" Number of inversions are %d \n", mergeSort(arr, 8));
```

This gives answer 28

^ | v • Reply • Share ›



Rahul → Guest • 7 months ago

change your array size to 8 from 5. It will work.

1 ^ | v • Reply • Share ›



MM → Rahul • 7 months ago

So what ?

^ | v • Reply • Share ›



helper bro • 9 months ago

we can use binary search tree for this perpose

2 ^ | v • Reply • Share ›



Kim Jong-il → helper bro • 9 months ago

It would be Better, you have written some line to support your claim.

3 ^ | v • Reply • Share ›



tarun • 9 months ago

@GeeksforGeeks: shouldn't all the element left in the left subarray in merge() be greater than every element in the right sub-array. That should also be added to inv_cnt, like this-

```
while (i <= mid - 1){
```

```
temp[k++] = arr[i++];
```

```
if(a[i] > a[j-1]) // This check is to avoid counting equal nos
```

```
inv_cnt += right-mid; // because this i is greater than all j
```

```
}
```

^ | v • Reply • Share ›



jayasurya_j → tarun • 4 months ago

I have the same doubt

^ | v • Reply • Share ›



Guest • 10 months ago

```
<script src="http://ideone.com/e.js/o3VlsE" type="text/javascript"></script>
```

^ | v • Reply • Share ›



Learner • 10 months ago

Inversion count will be sum of inversions in left-part, right-part

and number of inversions in merging

should the line given in code

```
inv_count = _mergeSort(arr, temp, left, mid);
```

Actually be

```
inv_count += _mergeSort(arr, temp, left, mid);
```

^ | v • Reply • Share ›



Udit • 10 months ago

because left and right subarrays are sorted, so all the remaining elements in left-subarray ($a[i+1]$, $a[i+2]$... $a[mid]$) will be greater than $a[j]$

at last it should be $b[j]$ instead of $a[j]$

1 ^ | v • Reply • Share ›



retard • a year ago

try YODANESS @spoj

^ | v • Reply • Share ›



Ayush Jain • a year ago

You can refer to my solution here in C and there is also a Java solution in the link below (by Marek Kirejczyk) which is much simpler and shorter than the above code

The logic is same but code is presented well and efficiently for understanding purpose

<http://stackoverflow.com/a/245...>

3 ^ | v • Reply • Share ›



Praveen Reddy • a year ago

Actually by using binary search tree we can know the number of inversions inserting an element if the element is less than root then we can easily count the number of nodes which are to the right side of the tree including the root or we can maintain an extra variable in the struct such that how many nodes are on its right and how many nodes are on its left

I think this will give $O(\log n)$ solution. found any different cases then comment and let me know please.. :)

^ | v • Reply • Share ›



ryan → Praveen Reddy • 8 months ago

no u can't ,try with 8,9,10,5,4,1,2,12

1 ^ | v • Reply • Share ›



Ayush Jain → Praveen Reddy • a year ago

This will be $O(n^2)$ as in the worst case you have to scan the entire right subtree For. e.g $arr[] = \{8, 7, 6, 5, 4, 3, 2, 1\}$

When you draw BST you have to scan all $n-1$ elements to increase their inversions which will result in $O(n^2)$

^ | v • Reply • Share ›

**kumar Saurabh** · a year ago

We can simply sort the array in $O(n \log n)$ and finally compare the original array with the sorted one. Count the number of places where the elements are different. The count will be inversion count.

^ | v · Reply · Share ›

**tweety** → kumar Saurabh · a year ago

no this method doesnt give inversion count. for the same example
 2,4,1,3,5 sorted array is
 1,2,3,4,5
 according to ur method inversion count should be 4 but it is 3

1 ^ | v · Reply · Share ›

**koder** → tweety · 5 months ago

For each element 'x' in the original array, find it's index in the sorted array, and add the difference of the indices. Note that this way we count twice each pair, so we just have to divide the final number by 2 to get the correct solution.

In the example above:

- 2: old_index = 0, new_index = 1, num_pairs = 1
- 4: old_index = 1, new_index = 3, num_pairs = 2
- 1: old_index = 2, new_index = 0, num_pairs = 2
- 3: old_index = 3, new_index = 2, num_pairs = 1
- 5: old_index = 4, new_index = 4, num_pairs = 0

- total = $(1 + 2 + 2 + 1 + 0)/2 = 3$

^ | v · Reply · Share ›

**rishabhjoshi** → tweety · a year ago

as per me the method by kumar saurabh can be corrected ,here it is giving 4 because he is not seeing the condition that $i < j$ so="" here="" we="" should="" see="" that="" too="" we="" can="" create="" a="" structure="" with="" first="" element="" as="" array="" element="" and="" second="" element="" as="" the="" array="" index="" .="" sort="" it="" as="" per="" elements="" and="" do="" check="" the="" condition="" of="" $i < j$ ="" while="" count++="" of="" inversion.="">

^ | v · Reply · Share ›

**tweety** → rishabhjoshi · a year ago

can u plz explain your algorithm in detail...??

what i have got from the above explanation is that u are first sorting your linked list on the basis of element's value.. after that u have to check index..how you are checking index and what is stored in i and j..???

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000

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I think It takes $O(n^2)$ bcoz for each value in...

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